

FORMULA SHEET

TRIGONOMETRY

PHYSICS

DRONA

JEE

FUNCTION RELATIONSHIPS

$$\begin{aligned}\sin \theta &= \frac{1}{\operatorname{cosec} \theta} & \operatorname{cosec} \theta &= \frac{1}{\sin \theta} \\ \cos \theta &= \frac{1}{\sec \theta} & \sec \theta &= \frac{1}{\cos \theta} \\ \tan \theta &= \frac{1}{\cot \theta} & \cot \theta &= \frac{1}{\tan \theta} \\ \tan \theta &= \frac{\sin \theta}{\cos \theta} & \cot \theta &= \frac{\cos \theta}{\sin \theta}\end{aligned}$$

PYTHAGOREAN IDENTITIES

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \\ \tan^2 \theta + 1 &= \sec^2 \theta \\ \cot^2 \theta + 1 &= \operatorname{cosec}^2 \theta\end{aligned}$$

ANGLE ADDITION FORMULAS

$$\begin{aligned}\sin (A+B) &= \sin A \cos B + \cos A \sin B \\ \sin (A-B) &= \sin A \cos B - \cos A \sin B \\ \cos (A+B) &= \cos A \cos B - \sin A \sin B \\ \cos (A-B) &= \cos A \cos B + \sin A \sin B \\ \tan (A+B) &= \frac{\tan A + \tan B}{1 - \tan A \tan B} \\ \tan (A-B) &= \frac{\tan A - \tan B}{1 + \tan A \tan B}\end{aligned}$$

DOUBLE ANGLE FORMULAS

$$\begin{aligned}\sin 2\theta &= 2 \sin \theta \cos \theta \\ \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ \cos 2\theta &= 1 - 2\sin^2 \theta \\ \cos 2\theta &= 2\cos^2 \theta - 1 \\ \tan 2\theta &= \frac{2 \tan \theta}{1 - \tan^2 \theta}\end{aligned}$$

TRIPLE ANGLE FORMULAS

$$\begin{aligned}\sin 3\theta &= 3 \sin \theta - 4 \sin^3 \theta \\ \cos 3\theta &= 4 \cos^3 \theta - 3 \cos \theta \\ \tan 3\theta &= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}\end{aligned}$$

HALF ANGLE FORMULAS

$$\begin{aligned}\sin \frac{\theta}{2} &= \pm \sqrt{\frac{1 - \cos \theta}{2}} \\ \cos \frac{\theta}{2} &= \pm \sqrt{\frac{1 + \cos \theta}{2}} \\ \tan \frac{\theta}{2} &= \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta}\end{aligned}$$

COFUNCTION FORMULAS (IN QUADRANT I)

$$\begin{aligned}\sin \theta &= \cos \left(\frac{\pi}{2} - \theta \right) & \cos \theta &= \sin \left(\frac{\pi}{2} - \theta \right) \\ \tan \theta &= \cot \left(\frac{\pi}{2} - \theta \right) & \cot \theta &= \tan \left(\frac{\pi}{2} - \theta \right) \\ \sec \theta &= \operatorname{cosec} \left(\frac{\pi}{2} - \theta \right) & \operatorname{cosec} \theta &= \sec \left(\frac{\pi}{2} - \theta \right)\end{aligned}$$

PRODUCT-TO-SUM FORMULAS

$$\begin{aligned}\sin A \cdot \sin B &= \frac{1}{2} [\cos (A-B) - \cos (A+B)] \\ \cos A \cdot \cos B &= \frac{1}{2} [\cos (A-B) + \cos (A+B)] \\ \sin A \cdot \cos B &= \frac{1}{2} [\sin (A+B) + \sin (A-B)] \\ \cos A \cdot \sin B &= \frac{1}{2} [\sin (A+B) - \sin (A-B)]\end{aligned}$$

SUM-TO-PRODUCT FORMULAS
$\sin A + \sin B = 2 \cdot \sin \left(\frac{A+B}{2} \right) \cdot \cos \left(\frac{A-B}{2} \right)$
$\sin A - \sin B = 2 \cdot \sin \left(\frac{A-B}{2} \right) \cdot \cos \left(\frac{A+B}{2} \right)$
$\cos A + \cos B = 2 \cdot \cos \left(\frac{A+B}{2} \right) \cdot \cos \left(\frac{A-B}{2} \right)$
$\cos A - \cos B = -2 \cdot \sin \left(\frac{A+B}{2} \right) \cdot \sin \left(\frac{A-B}{2} \right)$